Electrical machines and drives

Technical project report

Field-Oriented Control of a Squirrel-Cage Induction Motor

Indirect rotor-flux orientation and an SPWM–SVPWM comparison

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Simulation study

MATLAB / Simulink and Python

November 2026

Contents

1. Why use field-oriented control? 3

2. Control architecture 3

3. Clarke, Park, and indirect rotor-flux orientation 4

3.1 Clarke represents three phases as a stationary arrow 4

3.2 Park rotates the axes to follow the rotor flux 4

3.3 Estimate the flux magnitude, slip, and field angle 5

3.4 How the two current loops work together 5

4. Two ways to switch the same IGBT bridge 6

4.1 Sinusoidal PWM 6

4.2 Space-vector PWM and the 360 V active vectors 6

4.3 Seven-segment algorithm and worked example 7

5. Common machine and test conditions 10

6. Implementation and results 10

6.1 MATLAB and Simulink with fundamental blocks 10

6.2 MATLAB and Simscape Electrical with the library IGBT bridge 12

6.3 Python as an independent numerical implementation 17

7. Conclusion 19

Appendix A. Stationary-frame machine equations 19

Appendix B. Physical derivation of the PI gains 21

B.1 Current PIs on the d and q axes 21

B.2 Speed PI 22

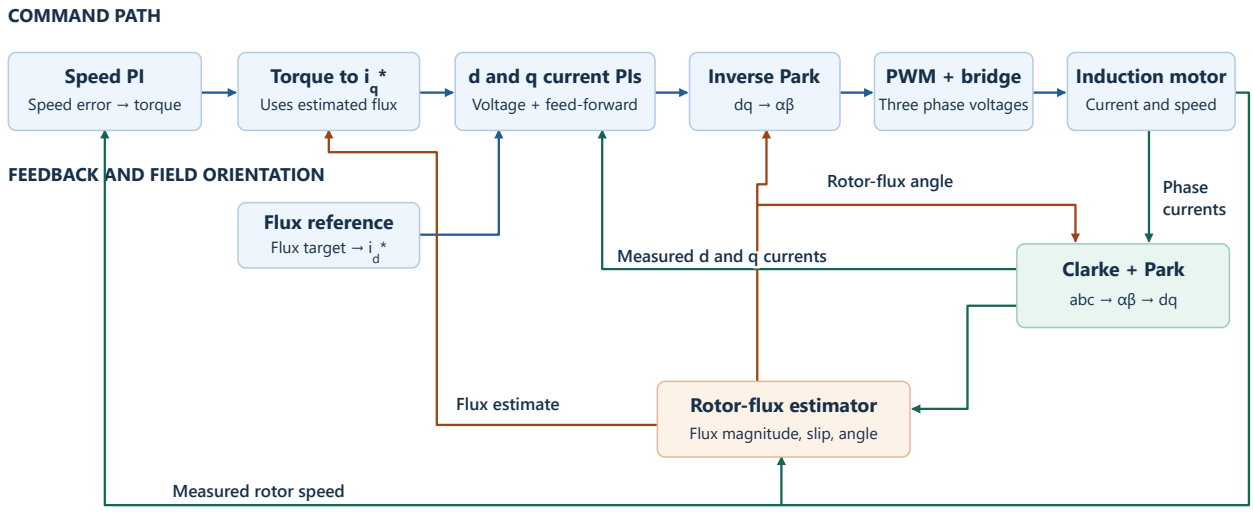
References 22

1. Why use field-oriented control?

An induction motor creates torque through the interaction of rotor flux and stator current. The three stator currents are alternating quantities, and the rotor flux rotates at a speed that differs from the mechanical rotor speed. Controlling the three phase currents directly makes it hard to see which part of the current builds the magnetic field and which part produces torque.

Field-oriented control (FOC) expresses the stator current in a reference frame that follows the rotor flux. The current along the flux axis, i_d, sets the rotor flux. The perpendicular current, i_q, sets the torque for a given flux. This gives the drive separate references for magnetization and torque, much like the field and armature currents of a separately excited DC machine. The speed controller can then request the torque needed to accelerate the rotor or reject a load disturbance.

2. Control architecture



The outer PI compares the speed reference and the motor-model speed, then outputs a torque request T_e^*. The controller converts it to a stator current reference. The flux target sets the other current reference.

$$i_q^* = \frac{T_e^*}{K_T \hat{\psi}_r}, \qquad K_T = \frac{3}{2} p \frac{L_m}{L_r}, \qquad i_d^* = \frac{\psi_r^*}{L_m}.$$

The drive uses $\psi_r^* = 0.8$ Wb, giving $i_d^* = 11.59$ A. The torque request is limited to ± 35 N m. The q-axis reference also respects the current budget left after magnetization, $|i_q^*| \le \sqrt{I_{\max}^2 - i_d^{*2}}$ with $I_{\max} = 25$ A. A minimum flux value protects the division during startup.

The two inner PIs compare i_d^*, i_q^* with measured i_d, i_q and produce v_d^*, v_q^* with feed-forward compensation. Inverse Park converts the voltage command to a stationary arrow, and the modulator and bridge apply the requested voltage to the machine.

3. Clarke, Park, and indirect rotor-flux orientation

3.1 Clarke represents three phases as a stationary arrow

The A, B, and C axes lie at $0°$, $120°$, and $240°$. Project each phase reading onto a fixed horizontal axis α and vertical axis β. Adding the projections gives a raw vector with $3/2$ of the phase amplitude for a balanced set. The $2/3$ normalization recovers the original amplitude.

$$\begin{bmatrix} x_\alpha \\ x_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 \end{bmatrix} \begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix}.$$

For example, $(10, -5, -5)$ A gives $(10, 0)$ A. The B and C vertical contributions cancel and their horizontal contributions add to A. For balanced three-wire currents, two coordinates fully describe the phase set. The same Clarke matrix removes a voltage offset shared by all three inverter legs.

3.2 Park rotates the axes to follow the rotor flux

Clarke's axes are stationary. Park uses θ_e, the electrical angle of the rotor-flux field, to rotate the coordinate axes. The d axis follows the flux and the q axis leads it by $90°$.

$$\begin{bmatrix} x_d \\ x_q \end{bmatrix} = \begin{bmatrix} \cos\theta_e & \sin\theta_e \\ -\sin\theta_e & \cos\theta_e \end{bmatrix} \begin{bmatrix} x_\alpha \\ x_\beta \end{bmatrix}, \qquad \begin{bmatrix} x_\alpha \\ x_\beta \end{bmatrix} = \begin{bmatrix} \cos\theta_e & -\sin\theta_e \\ \sin\theta_e & \cos\theta_e \end{bmatrix} \begin{bmatrix} x_d \\ x_q \end{bmatrix}.$$

If the flux angle is correct, the rotor flux reads $(\psi_{rd}, \psi_{rq}) = (\psi_r, 0)$. Stator current can still have both components. The aligned machine equations show their roles.

$$\tau_r \frac{d\psi_r}{dt} + \psi_r = L_m i_d, \qquad T_e = \frac{3}{2} p \frac{L_m}{L_r} \psi_r i_q, \qquad J \frac{d\omega_m}{dt} = T_e - T_L - B\omega_m.$$

The rotor flux changes gradually with i_d over the rotor time constant. At fixed flux, changing i_q changes torque. Speed changes when that torque differs from load and friction. If flux rises while the requested torque stays fixed, the torque-to-current conversion requests less i_q.

3.3 Estimate the flux magnitude, slip, and field angle

The indirect estimator combines measured stator current, machine parameters, and rotor speed from the plant model.

1. **Estimate the flux magnitude.** Filter i_d through the rotor lag, $\hat{\psi}_r(s) = L_m i_d(s)/(1 + \tau_r s)$. The rotor resistance R_r is a model parameter used in $\tau_r = L_r/R_r = 87.0$ ms.
2. **Calculate electrical slip.** Use the torque-producing current and the estimated flux, $\omega_{\text{sl}} = L_m i_q/(\tau_r \max(\hat{\psi}_r, \psi_{\min}))$. For the same flux, more i_q requires more slip.
3. **Update the field angle.** Add the electrical rotor speed $p\omega_m$ to slip, then integrate. $\theta_e = \int (p\omega_m + \omega_{\text{sl}})\, dt$. This angle describes the rotating rotor-flux field.

3.4 How the two current loops work together

Flux alignment separates the flux and torque relationships. The d and q voltage equations still contain speed-dependent coupling terms. The current controllers add feed-forward voltages that counter these terms.

$$v_{d,\text{ff}} = -\omega_e \sigma L_s i_q, \qquad v_{q,\text{ff}} = \omega_e \left(\sigma L_s i_d + \frac{L_m}{L_r} \hat{\psi}_r \right).$$

For example, raising i_q^* produces a q-axis voltage to raise i_q and a d-axis feed-forward correction to help the d-axis loop hold i_d. Correct flux orientation makes the two current readings meaningful. The current PIs and the feed-forward terms help each measured current follow its own reference.

Loop	Nominal bandwidth	Proportional gain	Integral gain
d- and q-axis current	400 Hz	9.91 V/A	3030 V/(A s)
Speed	30 Hz	1.056 N m s/rad	39.79 N m/rad

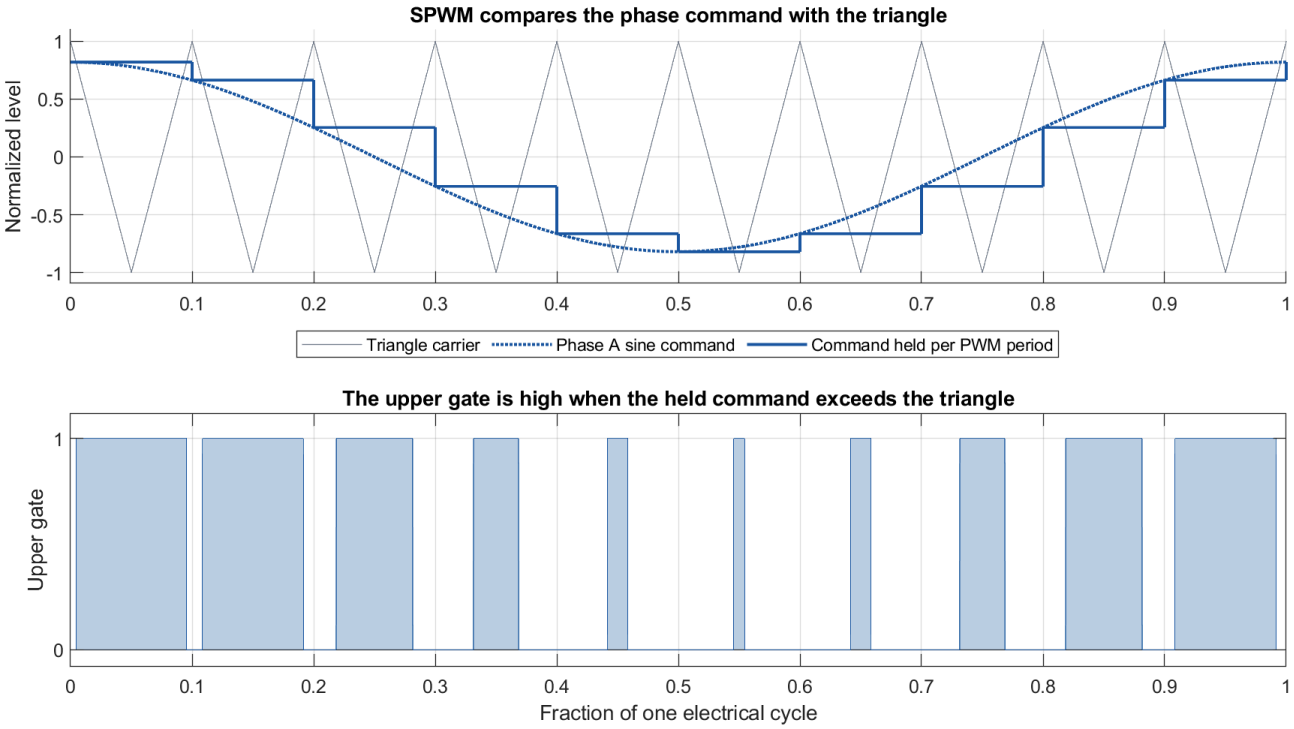
The current design uses the equivalent plant $1/(\sigma L_s s + R_\sigma)$, with $K_p = \sigma L_s \omega_{ci}$ and $K_i = R_\sigma \omega_{ci}$. For speed, $K_p = J\omega_{cs}$ and $K_i = K_p \omega_{cs}/5$. The inner loop is faster so the current can respond to torque requests before the outer loop changes them appreciably. The fundamental-block and Python controllers use back-calculation anti-windup. The Simscape controller uses discrete PI blocks with clamping.

4. Two ways to switch the same IGBT bridge

The three bits specify the upper gate commands for legs A, B, and C. A `1` connects the leg to the positive rail and a `0` to the negative rail in the ideal switching model. The lower gate is complementary. Both algorithms use a 10 kHz PWM period of 100 µs.

4.1 Sinusoidal PWM

Inverse Clarke gives phase-voltage commands from the stationary voltage arrow. The SPWM algorithm divides each by $V_{dc}/2$, holds each normalized reference for the PWM period, and compares it with a shared triangle that runs between -1 and $+1$. When a phase command exceeds the carrier, its upper gate goes high. A larger reference makes a wider pulse and a higher period-average pole voltage. The three references are separated by 120° in steady balanced operation.



The illustration shows ten carrier cycles per electrical cycle for readability. The simulated drive uses a 10 kHz carrier.

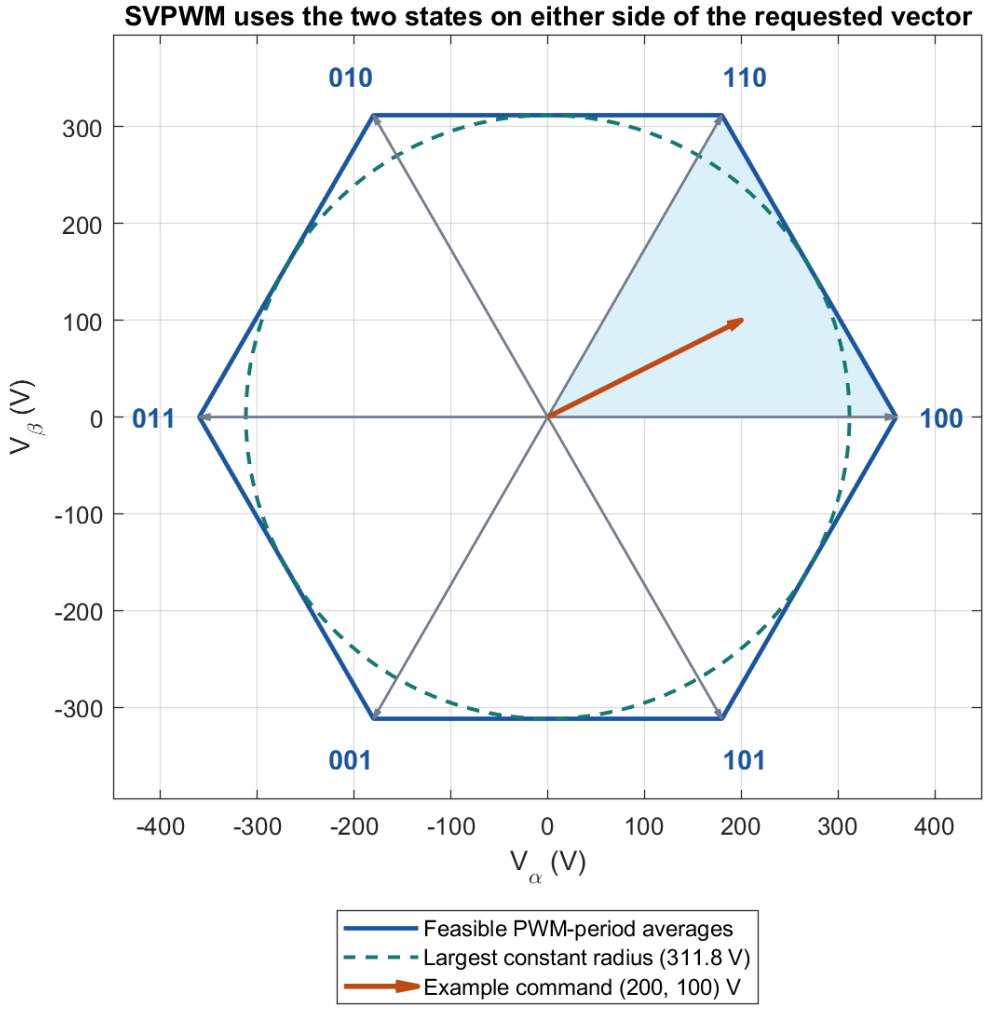
With the phase references applied directly, the linear phase-voltage peak is $V_{dc}/2 = 270$ V. MathWorks PWM reference

4.2 Space-vector PWM and the 360 V active vectors

For `100`, take the negative rail as 0 V. The terminal voltages are $(540, 0, 0)$ V. Applying Clarke gives the first active vector.

$$V_\alpha = \frac{V_{AB} + V_{AC}}{3} = \frac{540 + 540}{3} = 360\ \text{V}, \qquad V_\beta = \frac{V_{BC}}{\sqrt{3}} = 0.$$

For `110` the pole voltages are $(540, 540, 0)$ V, giving $(180, 311.8)$ V. Its magnitude is also 360 V and it lies at $60°$. Equivalently, its components are $360\cos 60°$ and $360\sin 60°$. The six active vectors are the corners of the voltage hexagon. The states `000` and `111` give zero line-to-line voltage because all three terminals share one rail.



The 360 V value is $2V_{dc}/3$, the largest α component at the `100` corner. At $30°$, the largest reachable arrow is the midpoint of the edge joining `100` and `110`, with magnitude $360\cos 30° = 311.8$ V. A constant-magnitude arrow must fit at every angle to produce balanced sine waves. Its linear limit is therefore $V_{dc}/\sqrt{3} = 311.8$ V, about 15.5% above the SPWM limit. Voltage hexagon and active vectors

4.3 Seven-segment algorithm and worked example

1. **Find the sector.** For the requested $(V_\alpha^*, V_\beta^*) = (200, 100)$ V, the angle is $26.6°$. It lies between `100` at $0°$ and `110` at $60°$.

2. **Calculate total time in each state.** Only `110` contributes to β here, so it takes $100/311.8$ of the period, or 32.08 µs. It also supplies about 57.7 V of average α voltage. State `100` supplies the remaining 142.3 V with 39.52 µs at 360 V. The two zero states share the remaining 28.41 µs.
3. **Place the times in order.** Start at `000`, visit the active state with one high bit, then the active state with two, then `111`. Retrace the order. The high-bit rule works in every sector. Split each active dwell across its two visits, and split the zero time equally between the two zero states.

State	`000`	`100`	`110`	`111`	`110`	`100`	`000`
Time in µs, rounded	7.10	19.76	16.04	14.20	16.04	19.76	7.10

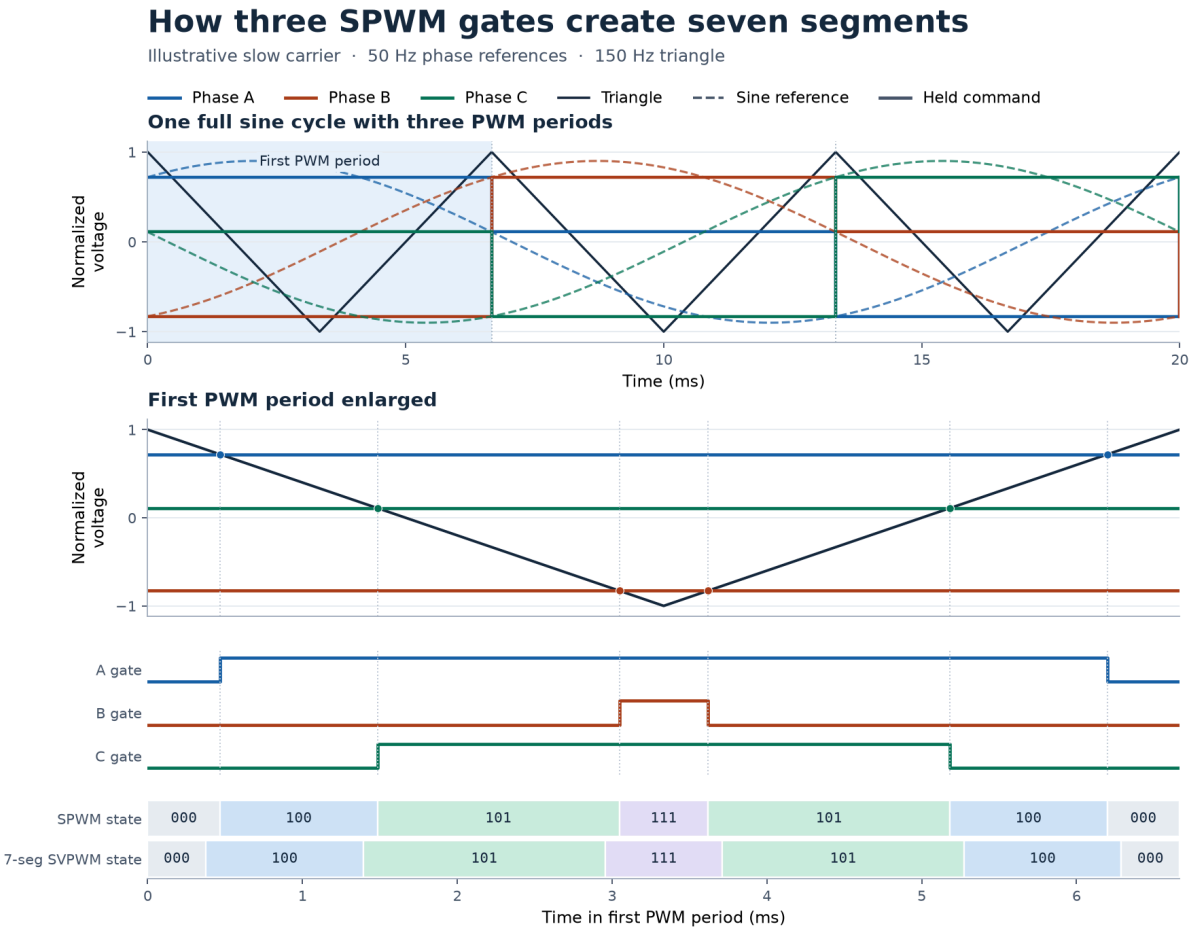
The general calculation uses the reference magnitude R_v, its angle δ measured from the start of the sector, and the period T_s.

$$T_1 = T_s \frac{\sqrt{3} R_v}{V_{dc}} \sin(60^\circ - \delta), \qquad T_2 = T_s \frac{\sqrt{3} R_v}{V_{dc}} \sin \delta, \qquad T_0 = T_s - T_1 - T_2.$$

If $T_1 + T_2 > T_s$, the requested voltage is outside the hexagon in that direction. Changing the switching frequency changes the absolute times but preserves the fraction of the period required. The implementation scales the two active times by the same factor when they exceed the period. TI space-vector PWM implementation

The fundamental-block and Python average-value models use an equivalent centered modulator. They add a common offset $-\frac{1}{2}(\max(v_a^*, v_b^*, v_c^*) + \min(v_a^*, v_b^*, v_c^*))$ to the three phase commands. This centers their highest and lowest values while preserving phase-to-phase voltage differences. The Simscape SVPWM model explicitly lays out the seven timed states. TI min-max implementation

Why SPWM can give the same seven segments. The teaching diagram uses 50 Hz sine references and a slowed 150 Hz triangle to fit three PWM periods in one electrical cycle.



In the enlarged period, the held commands are ordered A, C, B from highest to lowest. As the shared SPWM triangle falls, the three upper gates turn on in that order. As it rises, they turn off in reverse. Each leg has one continuous on pulse. Together, the three legs pass through `000` → `100` → `101` → `111` → `101` → `100` → `000`. Centered SVPWM schedules this order from the sector and dwell times. Its equivalent carrier adds a common offset, preserving the crossing order and shifting the crossing times.

Forward four-segment variant. The same dwell calculation can drive the sequence `000` → `100` → `101` → `111` in this sector. The two active states receive their full dwells and each zero state receives $T_0/2$. The next period restarts at `000`, so all three legs switch together at that boundary. Both zero states apply zero voltage between motor terminals.

5. Common machine and test conditions

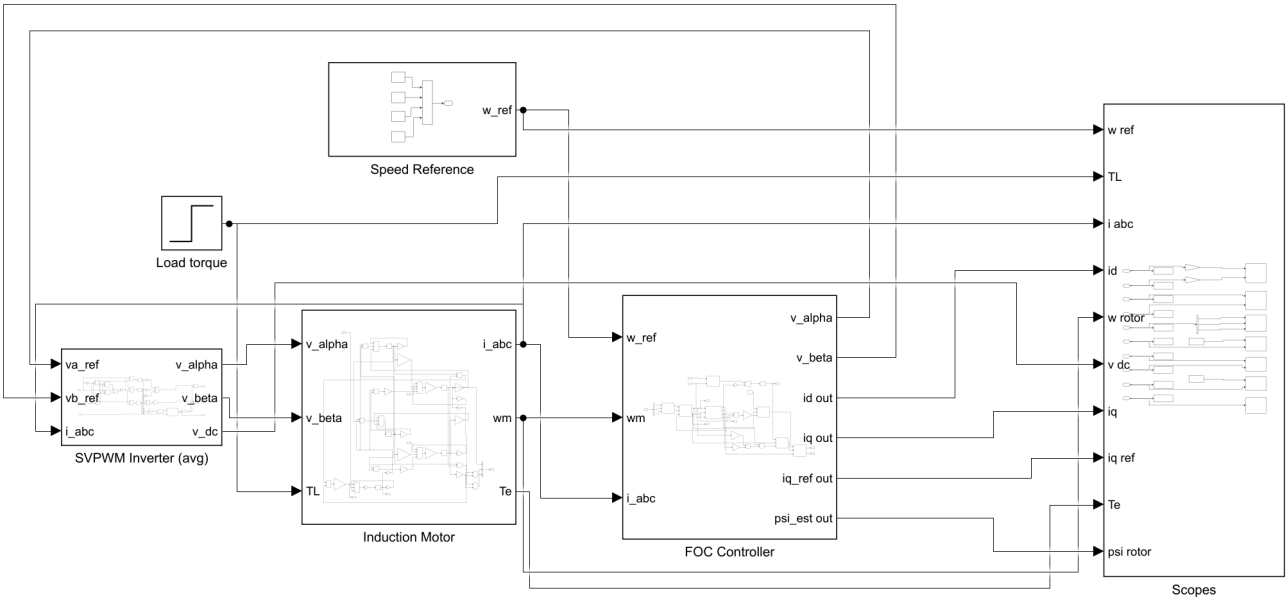
Parameter	Value	Parameter	Value
Rated power	5 kW	Rated line voltage	380 V RMS
Pole pairs p	2	Rated current	10.5 A
R_s	0.435 Ω	R_r	0.816 Ω
L_s, L_r	71 mH each	L_m	69 mH
Inertia J	0.0056 kg m²	Viscous friction B	0.0015 N m s/rad

The 2.4 s test starts at standstill and allows the flux to build. Speed steps reach 800, 1400, 600, and −800 rpm at 0.30, 0.80, 1.30, and 1.80 s. A positive 10 N m load torque is applied at 0.55 s and stays applied through the reversal. The three switching variants share the motor, bridge, controller, 540 V source, test conditions, and 1 µs electrical time step.

6. Implementation and results

6.1 MATLAB and Simulink with fundamental blocks

The first model implements the stationary-frame motor equations, PI controllers, reference transforms, and flux estimator using sums, gains, products, and integrators. The bridge uses the average pole voltages over a PWM period. The DC source drives the link through a series resistance and inductance, and a capacitor supplies the difference between source and bridge current.



The **Scopes** panel groups the live speed, torque, phase currents, d and q currents, rotor flux, and DC-link voltage. The speed, d and q currents, and flux views show the measured or estimated signal alongside its reference.

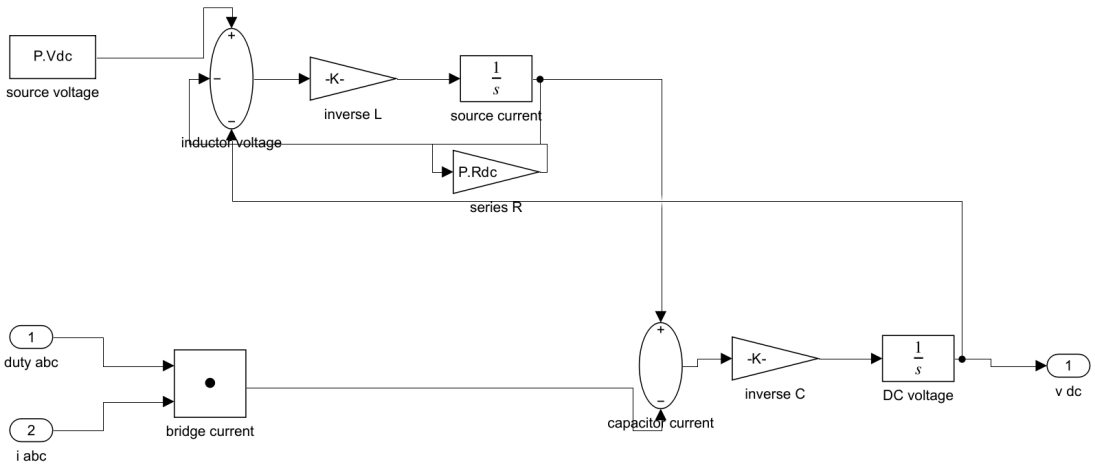
The modulator supplies the duty vector $d = [d_a, d_b, d_c]^\top$. For an ideal averaged bridge, $i_{\text{bridge}} = d^\top i_{abc}$ and $v_{\alpha\beta} = v_{dc} M d$, where M is the Clarke matrix. Thus the same command that changes the voltage applied to the motor changes the current drawn from the link.

$$\frac{di_s}{dt} = \frac{V_s - R_{\text{dc}} i_s - v_{dc}}{L_{\text{dc}}}, \qquad \frac{dv_{dc}}{dt} = \frac{i_s - d^\top i_{abc}}{C_{\text{dc}}}.$$

Here i_s is the DC source current. For the augmented electrical state $x = [i_s,\ v_{dc},\ x_{\text{IM}}^\top]^\top$, let $x_{\text{IM}} = [i_\alpha, i_\beta, \psi_{r\alpha}, \psi_{r\beta}]^\top$, $g = Md$, and let B_{IM} map the stationary voltage into the motor electrical equations. The command-dependent state matrix is

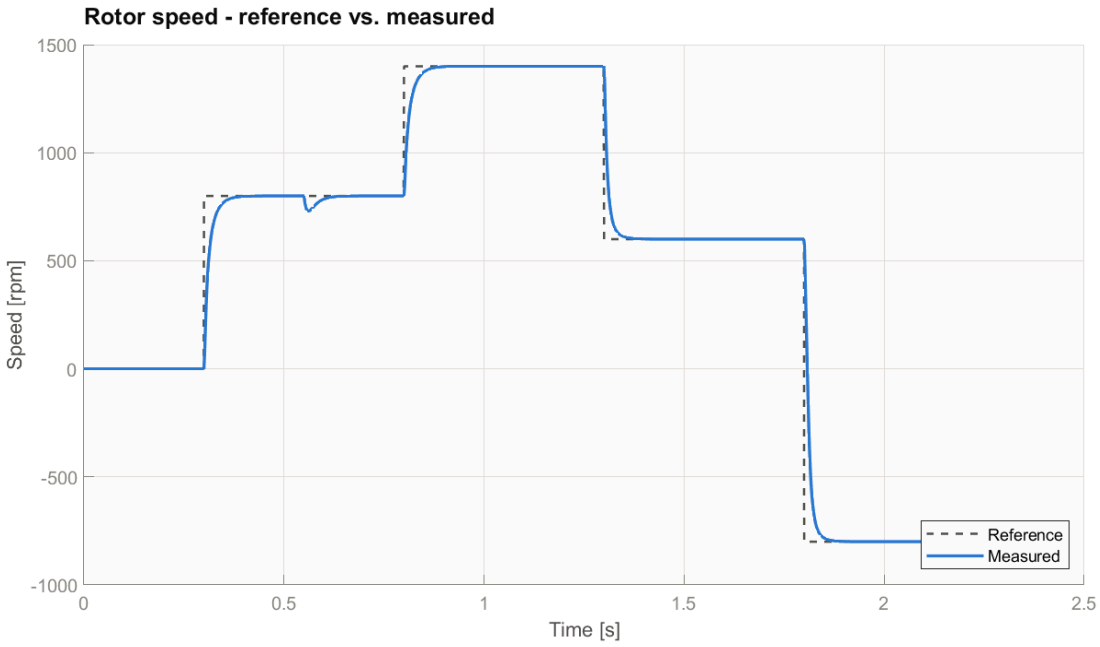
$$A(d, \omega_r) = \begin{bmatrix} -R_{\text{dc}}/L_{\text{dc}} & -1/L_{\text{dc}} & 0_{1\times 4} \\ 1/C_{\text{dc}} & 0 & -\dfrac{3}{2C_{\text{dc}}}[g^\top \quad 0 \quad 0] \\ 0_{4\times 1} & B_{\text{IM}} g & A_{\text{IM}}(\omega_r) \end{bmatrix}.$$

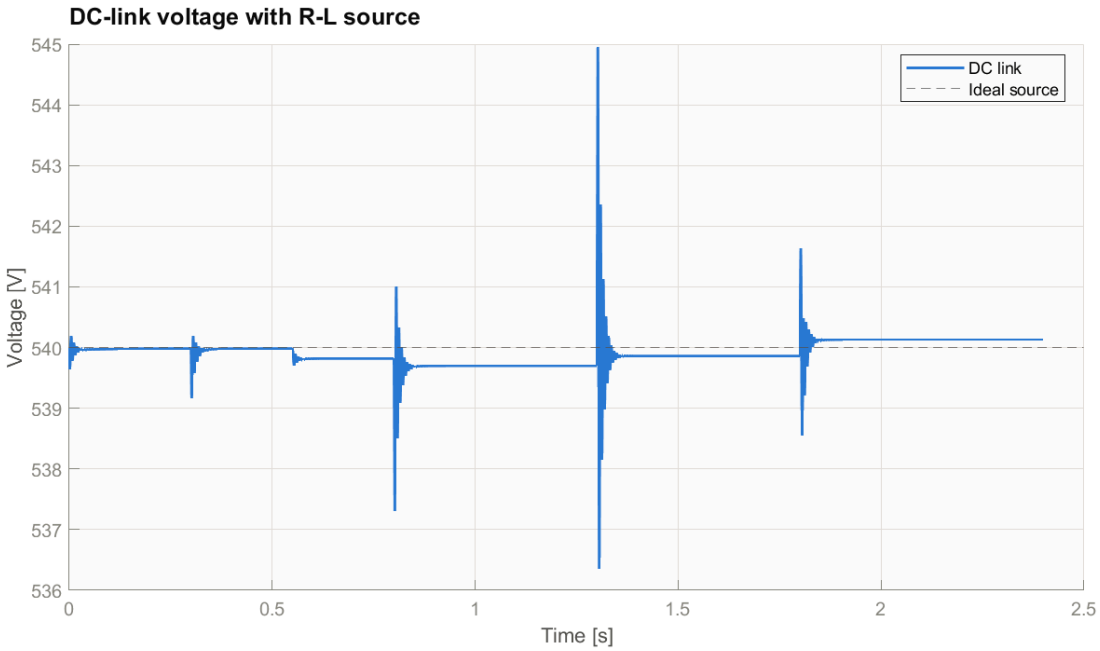
The upper-right coupling uses $d^\top i_{abc} = \frac{3}{2} g^\top i_{\alpha\beta}$ for the three-wire load. The lower-left voltage coupling uses $B_{\text{IM}} g$. The basic blocks evaluate these products directly. With a binary switching command, the same ideal bridge relationships describe the instantaneous topology. This version evaluates them with the average duty commands and the measured DC-link state.



The illustrative component values are $R_{\text{dc}} = 0.10\ \Omega$, $L_{\text{dc}} = 0.5$ mH, and $C_{\text{dc}} = 2.2$ mF, with a precharged 540 V capacitor. They are simulation assumptions. The run gives a link range of **536.4 to 544.9 V**, including transient sag and rise as power flows

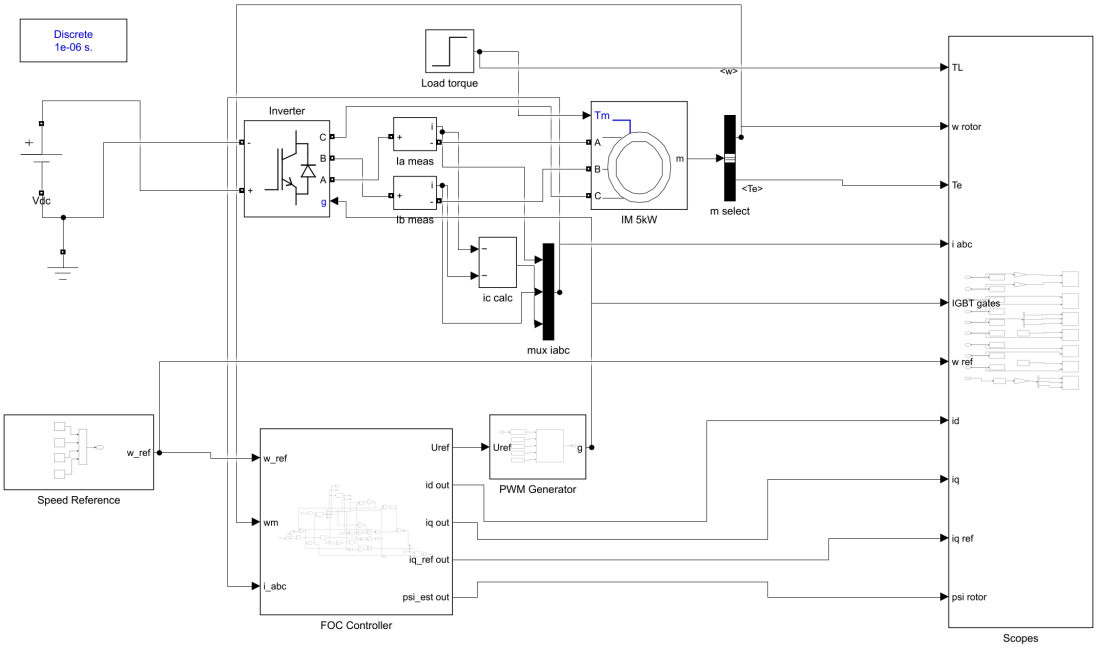
between the motor and the link. The speed steps settle in 70, 68, 52, and 59 ms within ±10 rpm. The load dip is 69.17 rpm and the final flux estimate is 0.800 Wb.



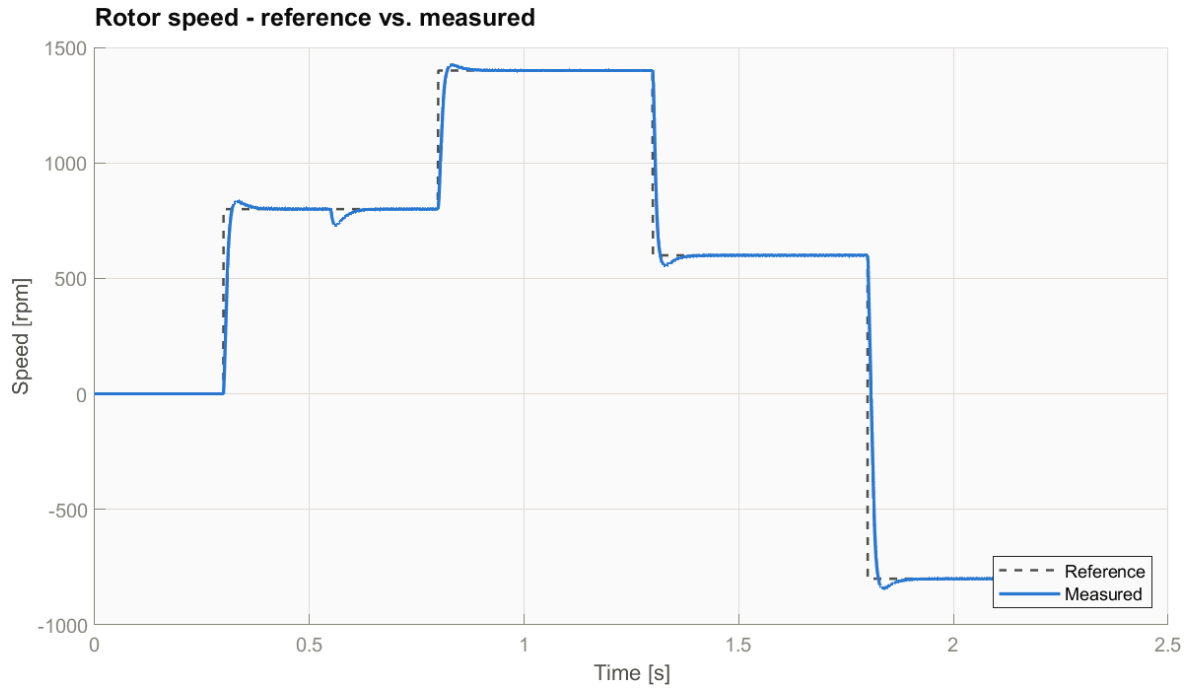


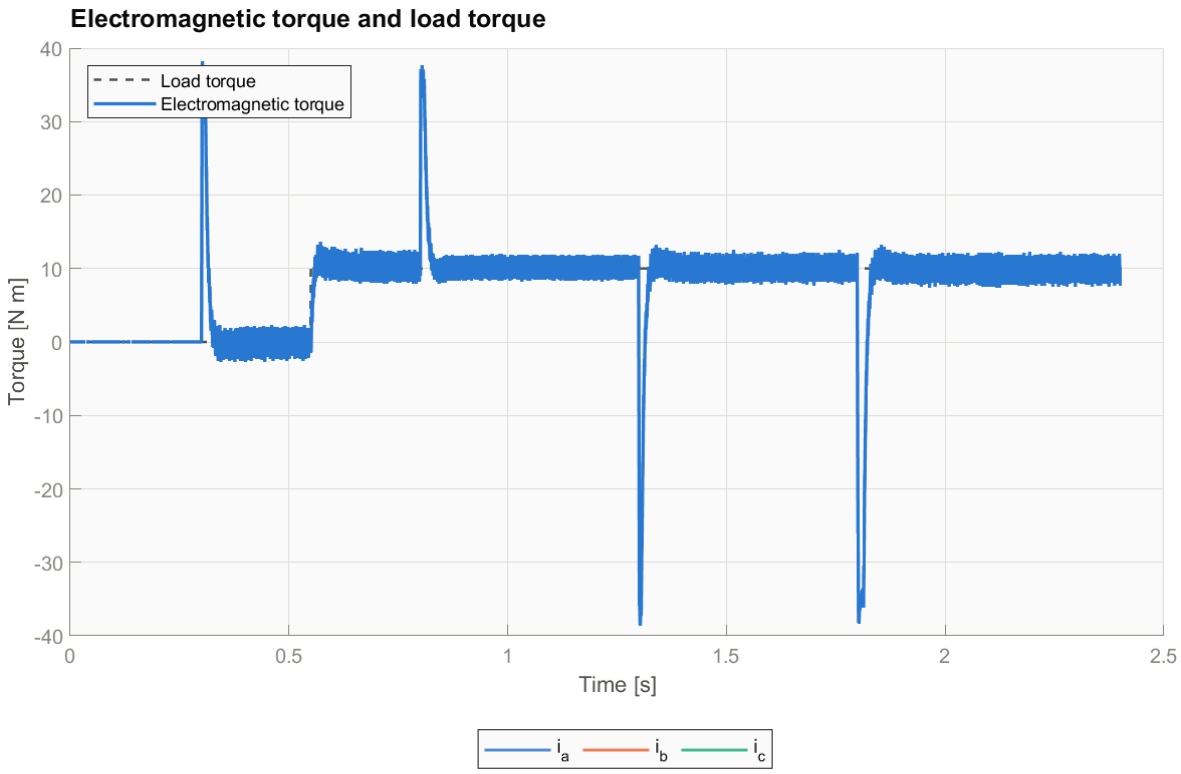
6.2 MATLAB and Simscape Electrical with the library IGBT bridge

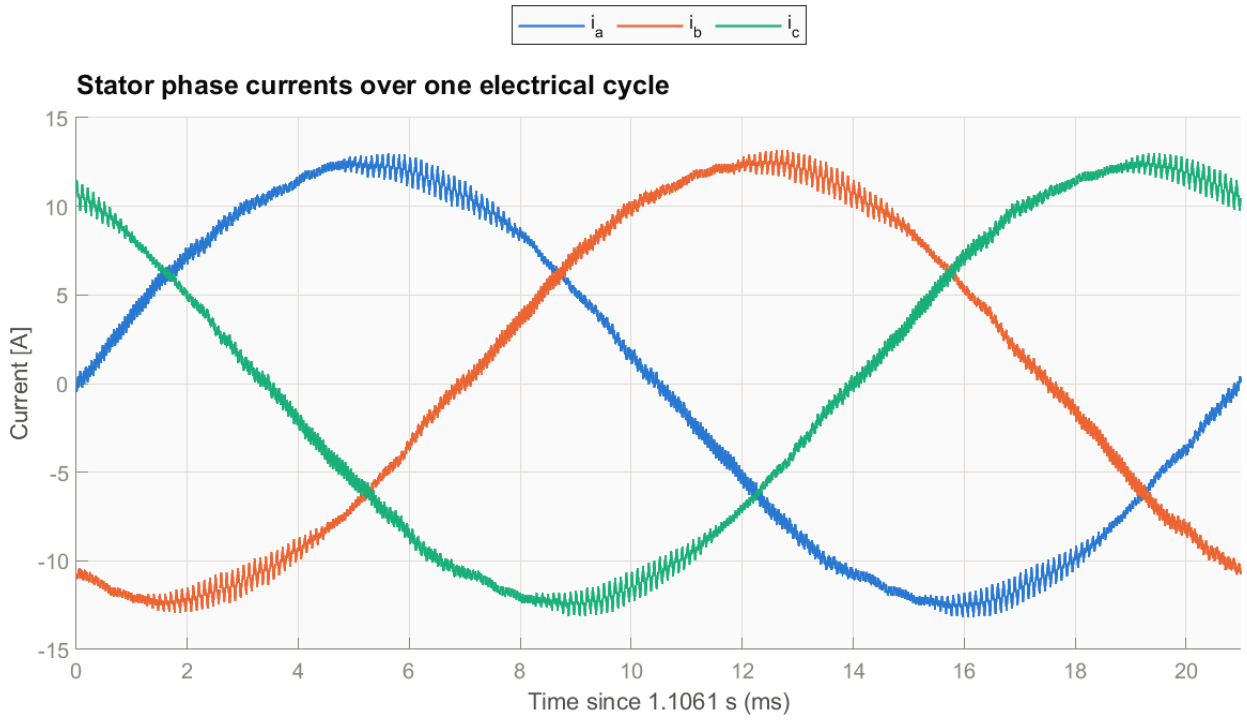
The switching model uses the Simscape Electrical induction-machine and Universal Bridge blocks with discrete controllers. It measures two phase currents and reconstructs the third as $i_c = -i_a - i_b$. The SPS Park blocks use a sine reference, so the controller adapts the axis order and sign to its cosine-referenced convention. The three graphical models share this drive. The SPWM variant uses a triangle and three comparators. The two SVPWM variants use the same sector and dwell calculations, with either the centered seven-segment or the forward four-segment sequence.

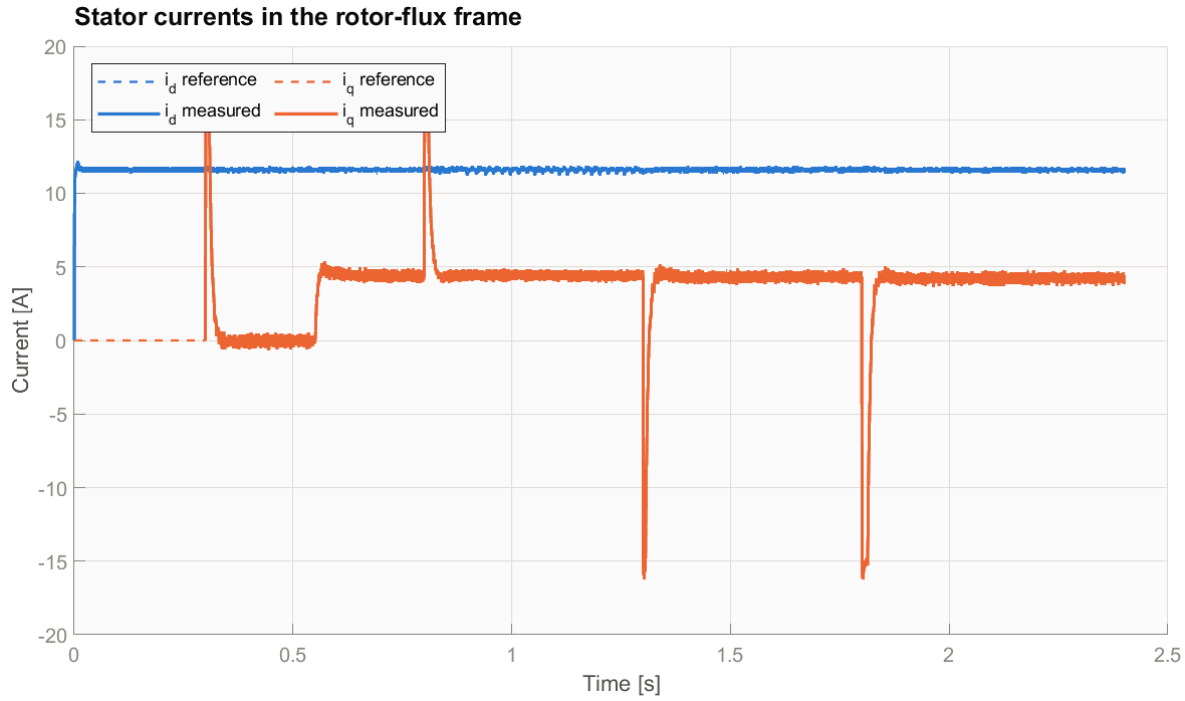


The **Scopes** panel shows the rotor speed, motor torque, phase currents, and currents in the rotor-flux frame. The following figures use the default seven-segment SVPWM run. The three phase currents share one set of axes and span one electrical cycle on the 1400 rpm plateau. The plot uses the original 1 µs samples. Phase C is reconstructed from the two measured currents.




Electromagnetic torque and load torque
Load torque
Electromagnetic torque
Torque [N m]
Time [s]
40
30
20
10
0
-10
-20
-30
-40
0
0.5
1
1.5
2
2.5

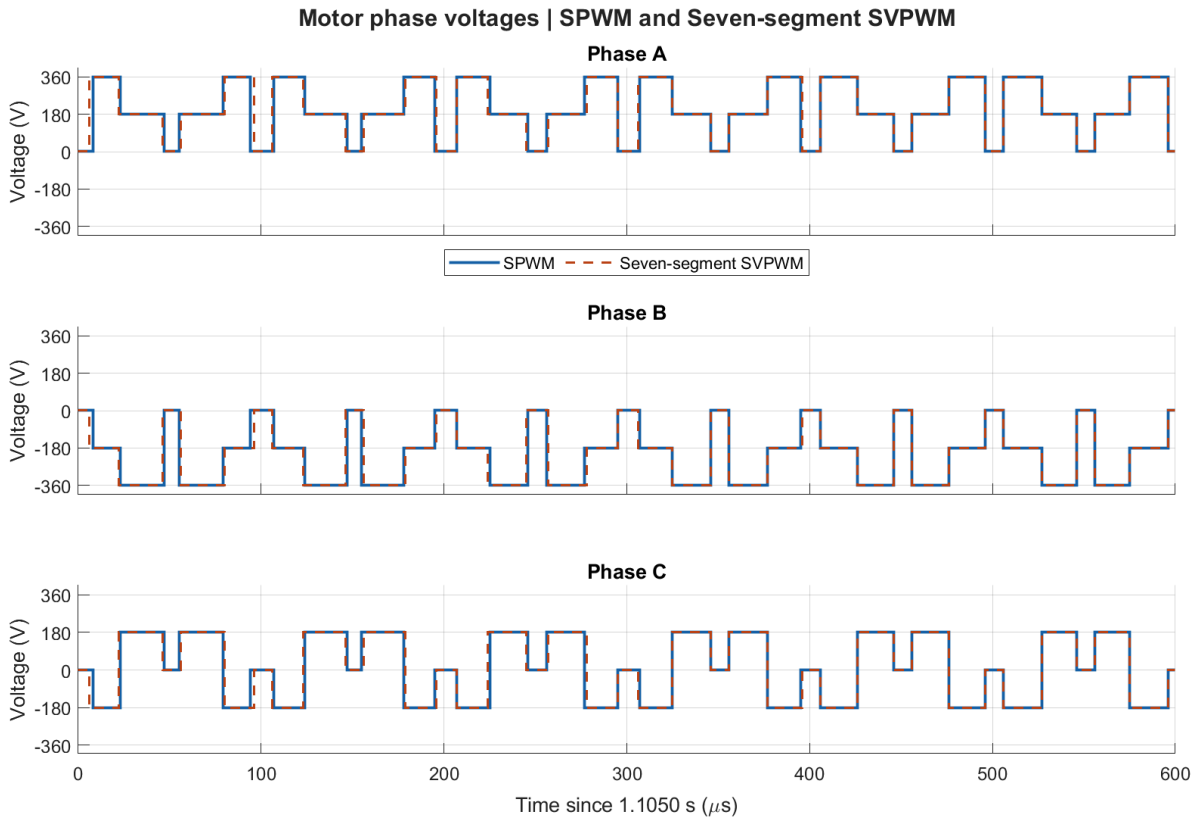

i_a
i_b
i_c
Stator phase currents over one electrical cycle
Current [A]
Time since 1.1061 s (ms)
15
10
5
0
-5
-10
-15
0
2
4
6
8
10
12
14
16
18
20

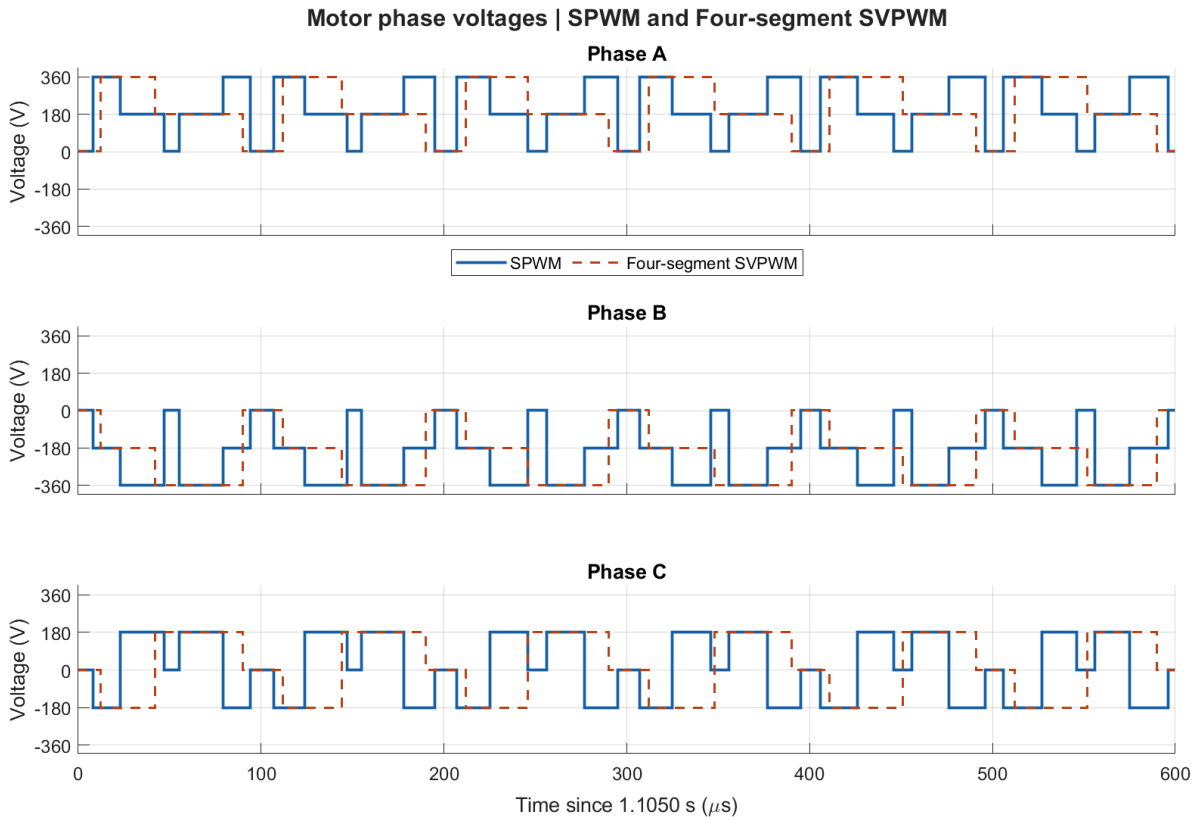


Switching voltage in the three graphical models. The three runs use the same 540 V DC bus, 10 kHz switching frequency, 1 µs electrical step and controller settings. The voltage and torque comparisons cover **1.1050 to 1.1056 s** at the 1400 rpm command. The phase voltages are relative to the virtual motor neutral and are derived from the three line voltages measured at the motor terminals.

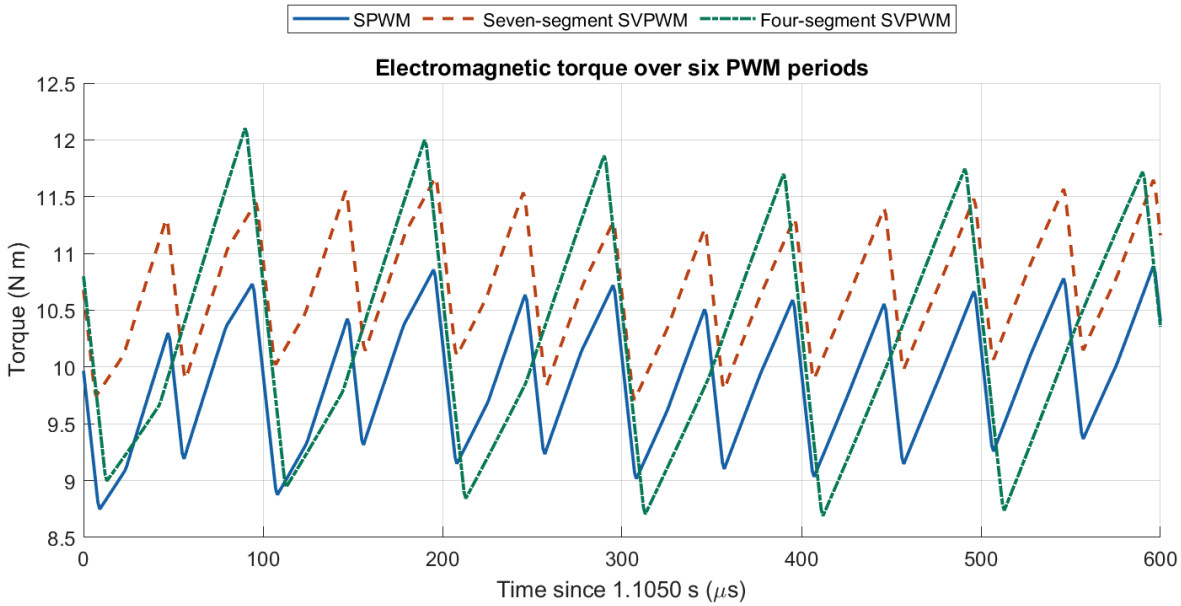
SPWM and seven-segment SVPWM. In the first period, SPWM spends 14 µs in `000` and 8 µs in `111`. In centered SVPWM, these two states last 10 µs each.

SPWM and four-segment SVPWM. Phase A averages about 194.4 V in both runs over the first period. In four-segment SVPWM, states `100` and `101` last 30 µs and 48 µs, respectively.


Motor phase voltages | SPWM and Seven-segment SVPWM
Phase A
Voltage (V)
360
180
0
-180
-360
SPWM
Seven-segment SVPWM
Phase B
Phase C
0
100
200
300
400
500
600
Time since 1.1050 s (μs)


Motor phase voltages | SPWM and Four-segment SVPWM
Phase A
Voltage (V)
360
180
0
-180
-360
SPWM
Four-segment SVPWM
Phase B
Phase C
0
100
200
300
400
500
600
Time since 1.1050 s (μs)

Electromagnetic torque over the same six periods. The four-segment run shows the largest raw torque swing in this window. Peak-to-peak values are calculated from the maximum and minimum of each recorded trace.

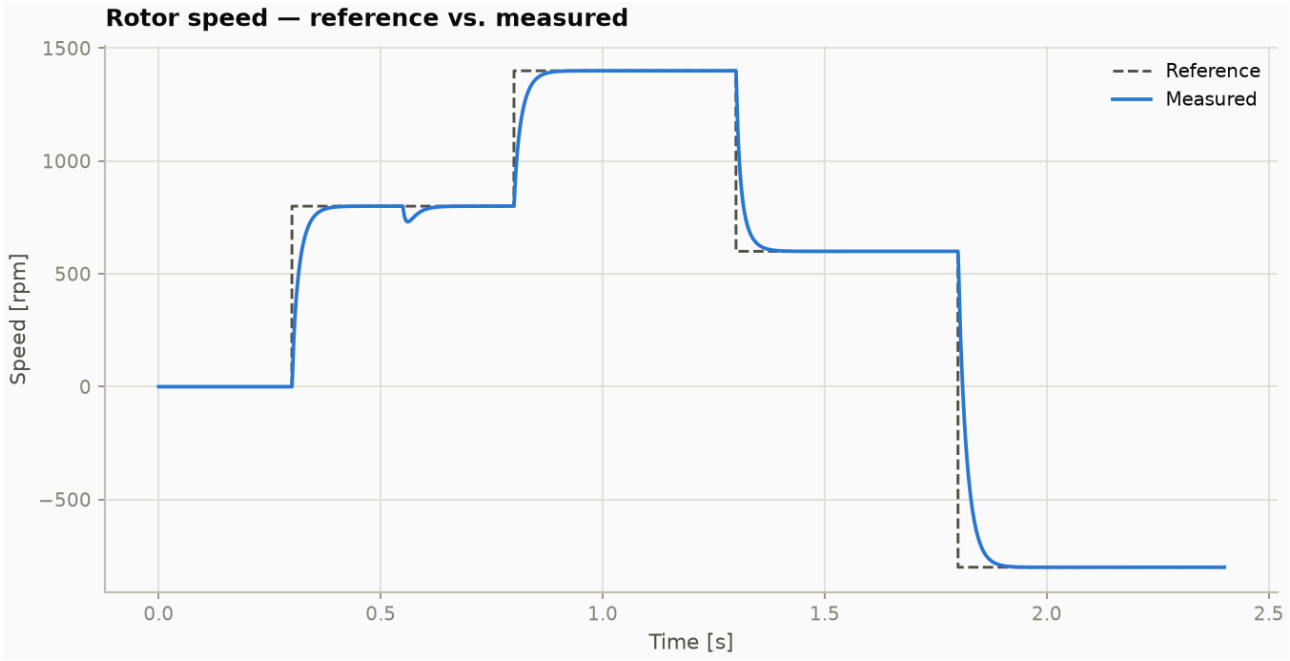


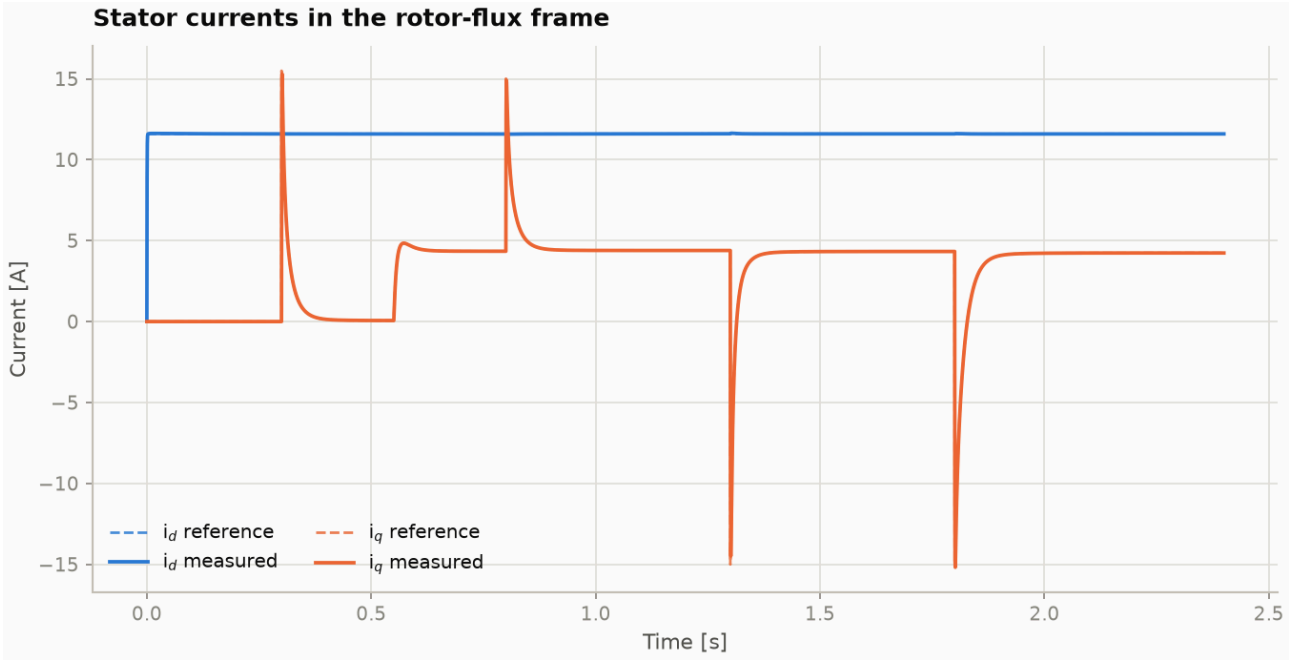
Modulation	Mean torque (N·m)	Peak-to-peak torque (N·m)
SPWM	9.87	2.14
Seven-segment SVPWM	10.69	1.96
Four-segment SVPWM	10.25	3.41

Run `compare_foc_pwm` in `matlab/simulink_library` to regenerate the voltage and torque plots and the underlying CSV data.

6.3 Python as an independent numerical implementation

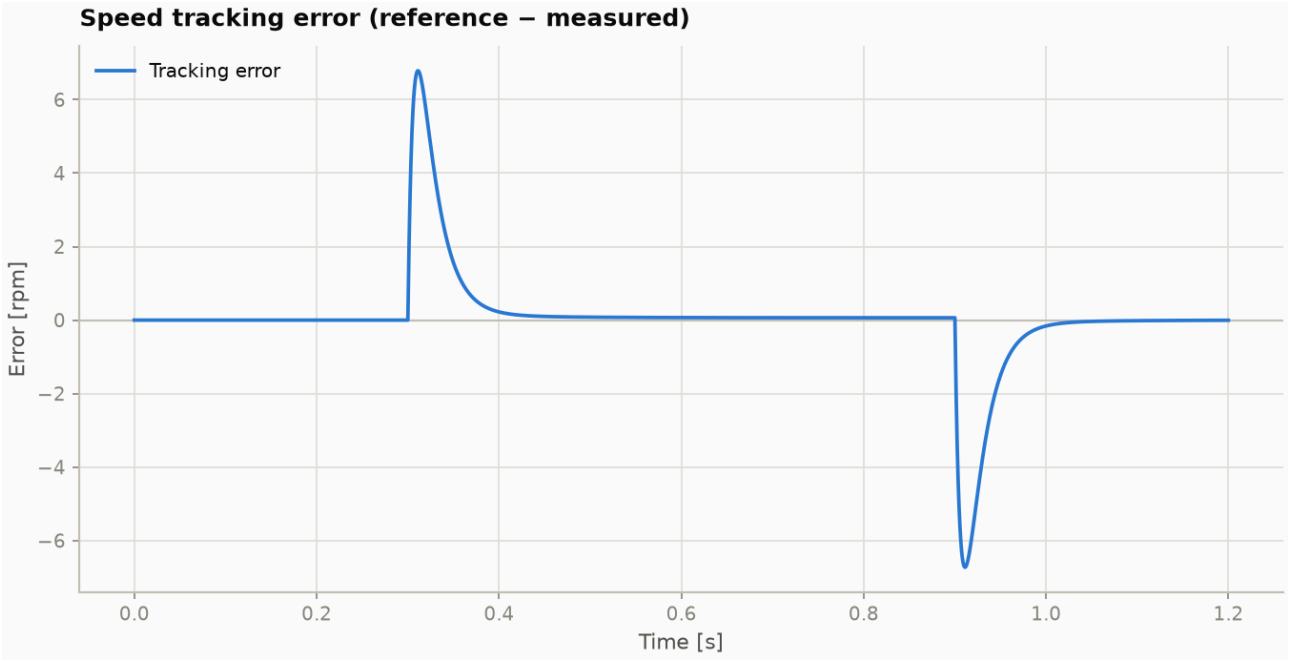
The third model tests how the same engineering study can be built in Python. A fourth-order Runge-Kutta integrator advances the stationary-frame motor equations. The controller, transforms, estimator, inverter, and plotting are separate modules. The default average-value inverter makes it practical to run scenarios quickly and inspect internal states such as both rotor-flux components.





In the common scenario, the four settling times are 79, 75, 71, and 95 ms, the load dip is 69.0 rpm, and the final rotor flux is 0.800 Wb. During the speed changes, i_q follows the torque demand while i_d returns near its magnetizing reference. The final q-axis rotor-flux residual is about 0.013 Wb.

The Python-only unloaded tracking study ramps to 1000 rpm from 0.30 to 0.90 s and holds 1000 rpm until 1.20 s. The maximum speed error is 0.08 rpm during the settled ramp window of 0.50 to 0.88 s and 0.16 rpm during the hold window of 1.00 to 1.20 s. The motor inertia contributes an integrator and the PI contributes another, which explains the small settled ramp error. The larger errors at the start and end of the ramp occur as the requested acceleration changes.



7. Conclusion

Shared step test	Python	Fundamental Simulink	Simscape SVPWM
Load-step dip (rpm)	69.0	69.17	69.70
Final rotor flux or estimate (Wb)	0.800	0.800	0.800
Peak phase current (A)	19.1	19.92	20.51

The three implementations reproduce the speed steps, reversal, and load rejection with the same rotor-flux target. The fundamental blocks expose the motor and DC-link equations. Python provides an independent numerical model and makes it easy to add speed profiles. The Simscape drive shows how the order and duration of the switching states shape phase voltage and electromagnetic torque. The flux reference stays fixed in this study, and the indirect orientation uses the nominal rotor time constant.

Appendix A. Stationary-frame machine equations

Write the stator current and rotor flux as two-vectors in the stationary frame and define $J_{90} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $\omega_r = p\omega_m$, $\sigma = 1 - L_m^2/(L_s L_r)$, and $R_\sigma = R_s + R_r(L_m/L_r)^2$. The electrical equations implemented by the fundamental-block and Python models are

$$\dot{\boldsymbol{\psi}}_r = -\frac{\boldsymbol{\psi}_r}{\tau_r} + \omega_r J_{90}\boldsymbol{\psi}_r + \frac{L_m}{\tau_r}\boldsymbol{i}_s,$$

$$\dot{\boldsymbol{i}}_s = \frac{1}{\sigma L_s}\left(\boldsymbol{v}_s - R_\sigma \boldsymbol{i}_s + \frac{L_m}{L_r \tau_r}\boldsymbol{\psi}_r - \frac{L_m}{L_r}\omega_r J_{90}\boldsymbol{\psi}_r\right).$$

$$T_e = \frac{3}{2} p \frac{L_m}{L_r}(\psi_{r\alpha} i_{s\beta} - \psi_{r\beta} i_{s\alpha}), \qquad J\dot{\omega}_m = T_e - T_L - B\omega_m.$$

This uses rotor flux as an electrical state. The equivalent current-state formulation follows from $\boldsymbol{\psi}_r = L_r \boldsymbol{i}_r + L_m \boldsymbol{i}_s$. The resulting transient inductance is $\sigma L_s = 3.94$ mH. The input matrix in section 6.1 is $B_{\mathrm{IM}} = [I_2/(\sigma L_s);\ 0_{2\times 2}]$.

Appendix B. Physical derivation of the PI gains

A PI regulator applies $u(t) = K_p e(t) + K_i \int_0^t e(\tau)\, d\tau$. The proportional term responds to the present error. The integral term builds the voltage or torque needed to remove a persistent error. In this drive, the two current PIs output voltages, and the speed PI outputs a torque request. The 400 Hz current-loop and 30 Hz speed-loop targets are design choices. The motor parameters determine the gains for those targets.

B.1 Current PIs on the d and q axes

From flux linkage to an electrical plant. The stator flux linkage is $\boldsymbol{\psi}_s = L_s \boldsymbol{i}_s + L_m \boldsymbol{i}_r$. Using $\boldsymbol{i}_r = (\boldsymbol{\psi}_r - L_m \boldsymbol{i}_s)/L_r$ separates the stator-current term from the rotor-flux term.

$$\boldsymbol{\psi}_s = L_\sigma \boldsymbol{i}_s + \frac{L_m}{L_r}\boldsymbol{\psi}_r, \qquad L_\sigma = L_s - \frac{L_m^2}{L_r} = \sigma L_s = 3.944\ \text{mH}.$$

In $v_s = R_s i_s + d\psi_s/dt$ on either stationary axis, the rotor-flux derivative from Appendix A contributes the current term $(L_m/L_r)(L_m/\tau_r) i_s$. Adding it to the stator resistance term gives

$$R_\sigma = R_s + \frac{L_m^2}{L_r \tau_r} = R_s + R_r \left(\frac{L_m}{L_r}\right)^2 = 1.206\ \Omega.$$

The inductance sets the voltage needed to change the current quickly. The resistance term combines the stator resistance with the rotor contribution reflected through the magnetic coupling. For tuning, the project uses the common nominal plant $G_i(s) = 1/(L_\sigma s + R_\sigma)$ for both axes. This isolates the fast electrical dynamics and treats the remaining flux and speed terms as disturbances or feed-forward terms. The simulation still retains the full machine and flux dynamics. Rotor-field-oriented current control

Place the zero and set the crossover. Write the PI as $C_i(s) = K_{p,i}(s + K_{i,i}/K_{p,i})/s$. Place its zero at the nominal electrical pole by setting $K_{i,i}/K_{p,i} = R_\sigma/L_\sigma$. This gives $C_i(s)G_i(s) = K_{p,i}/(L_\sigma s)$. Setting its gain to one at $\omega_{ci} = 2\pi \cdot 400 = 2513.3$ rad/s gives the two coefficients used by **each** current regulator.

$$K_{p,i} = L_\sigma \omega_{ci} = 9.91\ \text{V/A}, \qquad K_{i,i} = R_\sigma \omega_{ci} = 3030.19\ \text{V/(A s)}.$$

For a 1 A current error, the proportional term immediately requests about 9.91 V. If the error persists, the integral term continues building the voltage correction.

B.2 Speed PI

From torque to acceleration. The mechanical equation is $J\dot{\omega}_m = T_e - T_L - B\omega_m$. With the faster current loops assumed to deliver the requested torque, the plant from torque to speed is $G_\omega(s) = 1/(Js + B)$. At the 30 Hz design crossover, $\omega_{cs} = 188.5$ rad/s is much larger than $B/J = 0.268$ rad/s, so inertia dominates and $G_\omega(s) \simeq 1/(Js)$.

The proportional term supplies torque in proportion to speed error. Setting its nominal loop gain to one at ω_{cs} gives $K_{p,\omega} = J\omega_{cs}$. The design places the PI zero five times below crossover, at $\omega_z = \omega_{cs}/5 = 2\pi \cdot 6$ rad/s. Hence $K_{i,\omega} = K_{p,\omega}\omega_z$. This lets the proportional term dominate near crossover while the integral term builds the sustained torque needed to balance load and friction at zero speed error.

$$K_{p,\omega} = 0.0056(2\pi \cdot 30) = 1.056\ \mathrm{N\,m\,s/rad}, \qquad K_{i,\omega} = \frac{K_{p,\omega}\omega_{cs}}{5} = 39.79\ \mathrm{N\,m/rad}.$$

These are the nominal continuous-time tuning values used in the code. The discrete implementations update the regulators every 100 µs. The measured step responses also depend on the current and torque limits.

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